

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Helium nuclei / 2 protons and 2 neutrons (1) accept ${}^4_2\text{He}$ Emitted from the <u>nucleus</u> [at high speeds] (1)	2			2		
	(b)			$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{432} = 1.60 \times 10^{-3} \text{ yr}^{-1}$ or $5.08 \times 10^{-11} \text{ s}^{-1}$ (1) Either: Substitution: $A = A_0 \exp(-(1.60 \times 10^{-3})(30))$ (1) $\frac{A}{A_0} 100\% = 95.3\%$ (1) % decrease in activity = $100 - 95.3 = 4.7\%$ (1) or: 30 years $\ll$ $\frac{1}{2}$ -life [or by impl.] (1) $\therefore$ Fractional decay in 30 years = $30\lambda = 0.048$ (1) $\therefore$ % decrease in activity = 4.8% (1) Accept 5% Alternative: 30 years is $30/432 = 0.07$ half-lives (1) $A = \frac{1}{2}^{0.07}$ (1) $= 0.95$ (1) So activity has decreased by 5% (1)	1	1 1 1		4	3	
	(c)			Any 2 $\times$ (1) from: - Alpha particles pose little risk unless inhaled or ingested. - Very few (or none) make it to the outside of the detector. - If they do, they are stopped by a few cm of air and are unable to penetrate surface of skin. To award both marks there must be a conclusion present i.e. a minimum of no stated.			2	2		